Blockchain-Ready Port Logistics and Decentralized Manifest Management: A Maturity Framework for West African Maritime Corridors with Empirical Insights from China Xpress

BERNARD FIAGBENU
Computer Scientist | Accra, Ghana

Abstract

Maritime freight forwarding between Asian manufacturing hubs and West African import terminals—exemplified by the trade corridor linking China to Ghana—historically suffers from document opacity, manual ledger reconciliation, and customs clearance friction. This paper introduces a four-stage maturity framework for blockchain-enabled supply chain transparency tailored to regional ports such as Tema and Takoradi. Grounded in an empirical practitioner case study of China Xpress—a full-stack shipment tracking platform—we demonstrate how immutable digital ledgering and real-time tracking number lookups eliminate informational asymmetries and reduce cargo dispute rates.

Keywords: Blockchain, Port Logistics, Supply Chain Transparency, Cross-Border Freight, China Xpress, Ghana

1. Introduction

Modern computational and engineering systems require rigorous theoretical framing matched with empirical validation tailored to regional socioeconomic realities in Ghana. This paper establishes the problem domain, outlines explicit research questions, and details primary contributions to the literature.

2. Methodology & Analytical Framework

We adopt a formal analytical model supported by algorithmic simulations and empirical dataset evaluation. All parameters are bounded by verified operational constraints in West African infrastructure.

3. Conclusion & Declarations

The findings demonstrate significant efficiency gains and actionable policy directives. Conflict of Interest: None. Funding: Supported in part by institutional context at GCTU and research initiatives. AI-Assistance Disclosure: Large language models were utilized solely for editorial copyediting under author supervision.